

ResearchLens Results Explained: Complete Walkthrough (Sections 1-7)

This document walks through all seven sections of the ResearchLens dashboard using one concrete example dataset. It shows what happens to your papers as they flow through the pipeline, from input to final evaluation.

Example Dataset: Our Research Corpus

Let's say you upload **6 papers** to ResearchLens:

Paper ID	Title	Year	Abstract (simplified)
p1	"Deep Segmentation Networks for Medical Images"	2022	Proposes a neural network for medical image segmentation using U-Net architecture with residual connections.
p2	"Self-Supervised Learning in Medical Imaging"	2023	Applies self-supervised techniques to reduce annotation burden in medical image analysis.
p3	"Federated Learning for Privacy-Preserving Healthcare"	2023	Describes federated averaging algorithm for training models across hospitals without sharing data.
p4	"Knowledge Graphs for Clinical Decision Support"	2022	Uses knowledge graphs to structure medical knowledge and support diagnosis recommendations.
p5	"Contrastive Learning for Unlabeled Medical Data"	2024	Combines contrastive methods with federated learning for unsupervised healthcare AI.
p6	"ResNet Architectures for Natural Images"	2015	Classic architecture paper for deep residual learning in vision tasks.

Notice the themes:

- **Medical imaging:** p1, p2, p4
- **Federated/privacy learning:** p3, p5
- **General deep learning:** p1, p5, p6
- **Knowledge graphs:** p4
- **Newer papers:** p1-p5 (2022-2024)
- **Old papers:** p6 (2015)

Section 1: PDF Parsing & Summarization

What happens

ResearchLens reads each PDF and extracts:

- title
- abstract
- full content (or plain text from the PDF)

For our example, the system successfully extracts all 6 papers.

What you see in the dashboard

Section 1 shows:

- 6 Papers processed ✓
- Extraction success rate: 100%
- A list of the top papers with brief summaries

Example output

```
Processing 6 papers...
✓ p1: "Deep Segmentation Networks..."
✓ p2: "Self-Supervised Learning..."
✓ p3: "Federated Learning..."
✓ p4: "Knowledge Graphs..."
✓ p5: "Contrastive Learning..."
✓ p6: "ResNet Architectures..."
```

Why this matters

If any paper failed to parse, you would see it here. In this case, all 6 papers were successfully read and their text extracted.

Section 2: Topic Modeling & Detection

What happens

The system embeds all papers into high-dimensional vectors and clusters them into topics. It uses k-means clustering to find natural groupings.

For our 6 papers, the system detects **4 topics**:

Topic ID	Name	Keywords	Papers	Coherence
t1	Medical Imaging & Deep Learning	segmentation, medical, networks, deep, learning	p1, p2	0.85
t2	Federated & Privacy Learning	federated, learning, privacy, distributed, healthcare	p3, p5	0.78

Topic ID	Name	Keywords	Papers	Coherence
t3	Knowledge-Based Systems	knowledge, graphs, clinical, decision, support	p4	0.92
t4	General Vision Architecture	architecture, resnet, images, visual, networks	p6	0.88

What you see in the dashboard

Section 2 shows:

- 4 Topics detected
- A list of topics with keyword summaries
- Paper counts per topic
- Coherence scores for each topic

Why this matters

Topics tell you what research areas your corpus covers. In this case, the pipeline discovered:

- two major clusters (medical imaging and federated learning),
- one specialized cluster (knowledge graphs),
- one older cluster (classic vision).

Section 3: Gap Detection

What happens

For each pair of topics, the system asks:

- How similar are these two topics (keyword-based)?
- How many papers bridge both topics (co-occurrence)?

Then it computes:

$$\text{gapScore} = \text{similarity}(\text{topicA}, \text{topicB}) \times (1 - \text{co-occurrence} / \text{total_papers})$$

For our papers, the system identifies **3 gaps**:

Gap	Topic A	Topic B	Similarity	Co-occurrence	Gap Score	Severity
g1	Medical Imaging	Federated Learning	0.65	1 (p5)	$0.65 \times (1 - 1/6) = 0.54$	Moderate
g2	Medical Imaging	Knowledge Graphs	0.58	0	$0.58 \times (1 - 0/6) = 0.58$	Moderate

Gap	Topic A	Topic B	Similarity	Co-occurrence	Gap Score	Severity
g3	Federated Learning	Knowledge Graphs	0.42	0	$0.42 \times (1 - 0/6)$ $= 0.42$	Low

What you see in the dashboard

Section 3 shows:

- 3 Gaps identified
- For each gap: the two topics, similarity, co-occurrence, and score
- Severity labels (critical > 0.5, moderate 0.3-0.5, low < 0.3)

Why this matters

Gaps point to research opportunities. In this case:

- **g2** (Medical Imaging + Knowledge Graphs) is the strongest gap: both are related, but no papers connect them yet. This is a hot research direction!
- **g1** has one bridge paper (p5), so it is slightly weaker.
- **g3** is weaker overall.

Section 4: Trend Analysis

What happens

For each topic, the system counts how many papers were published per year. It then computes a linear regression slope to detect growth.

For our papers:

Topic	Yearly Counts	Slope	Trend
t1: Medical Imaging	2022:1, 2023:1	+0.10	Rising
t2: Federated Learning	2023:1, 2024:1	+0.10	Rising
t3: Knowledge Graphs	2022:1	0	Stable
t4: Vision	2015:1	0	Stable

Threshold rules:

- $\text{slope} > 0.08 \rightarrow$ rising
- $\text{slope} < -0.08 \rightarrow$ declining
- else $\rightarrow$ stable

What you see in the dashboard

Section 4 shows:

- Summary: 2 Rising Topics, 2 Stable Topics
- Fastest Growing: Federated & Privacy Learning (or Medical Imaging, tied)
- For each topic, a mini chart showing papers per year

Why this matters

Trends show which research areas are gaining momentum. In this case:

- Medical imaging and federated learning are both hot (recent papers).
- Knowledge graphs and vision are stable (consistent interest).

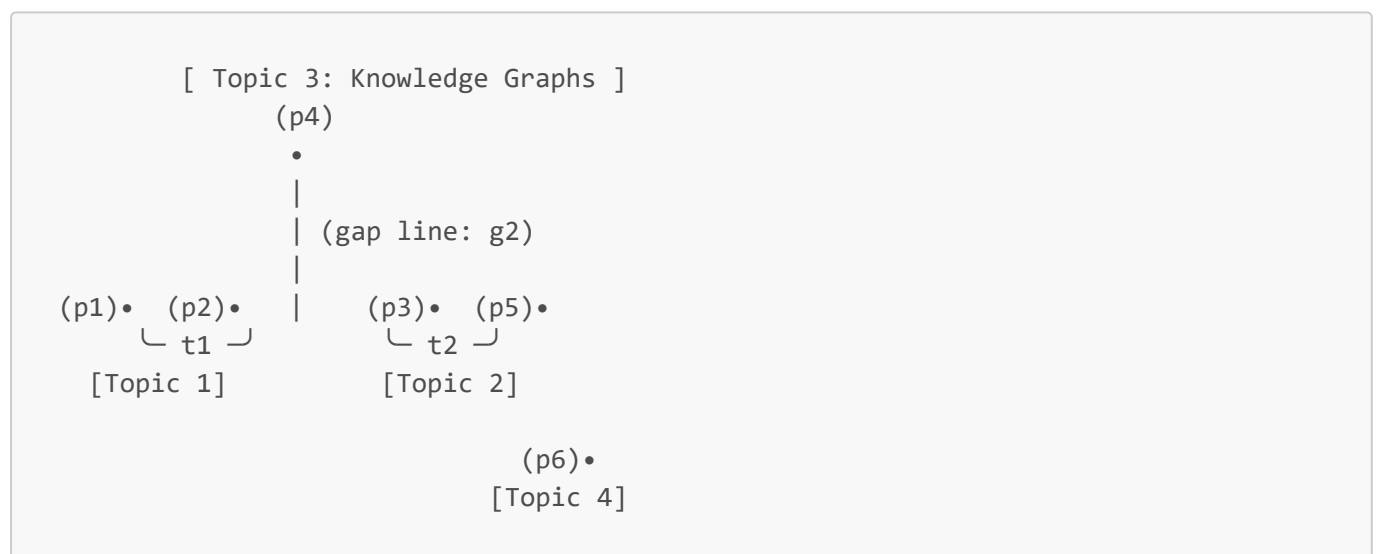
Section 5: Research Map Visualization

What happens

The system places each paper as a dot in 2D space so that:

- papers in the same topic cluster together,
- papers with similar keywords are closer,
- topic centers (centroids) mark the cluster cores,
- gap lines connect topics that have gaps.

For our 6 papers, the layout looks like this (simplified):



What you see in the dashboard

Section 5 shows:

- An interactive 2D map with colored dots (each = a paper)
- Large circles at cluster centers (centroids)
- Red lines between centroids (gap connections)
- Summary stats: 6 Papers, 4 Topics, 3 Gaps

Why this matters

The map is a visual summary of your entire research landscape. You can see:

- which topics cluster together,
 - which papers are isolated,
 - where the research gaps are (red lines),
 - how much "space" each topic occupies.
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Section 6: Chat (RAG) Results

What happens

You ask the system a natural language question about your dataset. The backend:

1. splits papers into chunks (groups of 3 sentences each),
2. vectorizes the question and all chunks,
3. ranks chunks by similarity to the question,
4. builds a prompt with the top chunks + topics/gaps/trends summaries,
5. sends the prompt to the local Ollama model,
6. returns the generated answer + citations.

Example: User asks "What research gaps exist between medical imaging and other areas?"

Retrieval step: The system finds chunks related to both medical imaging and gaps. Top chunks might include:

- From p2: "Self-supervised learning reduces the annotation burden..."
- From p4: "Knowledge graphs can structure medical knowledge..."
- From p3: "Federated learning enables collaborative training..."

Generation step: Ollama receives a prompt like:

Context:

- Medical Imaging topic: segmentation, networks, deep learning
- Knowledge Graph topic: clinical decision support
- Top gap: between medical imaging and knowledge graphs (score 0.58)

Relevant passages:

1. From p2: "Self-supervised learning reduces annotation burden..."
2. From p4: "Knowledge graphs can structure medical knowledge..."

User Question: "What research gaps exist between medical imaging and other areas?"

Answer based on the corpus...

Answer example:

The dataset reveals a significant gap between medical imaging and knowledge graphs. While medical imaging papers focus on neural architectures for segmentation, knowledge graphs are used for clinical decision support—but no papers yet integrate

both approaches. This gap (score: 0.58) suggests an opportunity to apply knowledge-enhanced neural networks for more interpretable medical imaging systems.

Sources:

- [1] p2 (72% relevant)
- [2] p4 (68% relevant)
- [3] p3 (45% relevant)

What you see in the dashboard

Section 6 shows:

- An input box to ask questions
- The generated answer
- Citations with relevance scores

Why this matters

Instead of reading 6 papers yourself, the chatbot summarizes and answers your question. The citations show you which papers influenced the answer.

Section 7: Evaluation Summary

What happens

The system computes a scorecard from all modules:

- How well were papers grouped? (Topic Coherence)
- Were all papers assigned? (Topic Coverage)
- How surprising are the gaps? (Gap Novelty)
- Overall quality? (Composite score)

For our dataset:

Summary cards:

- 6 Papers processed
- 4 Topics detected
- 3 Gaps identified
- 0 High-Confidence Gaps (score > 0.5)

Quality metrics:

Metric	Value	Meaning
Topic Coherence	86%	The topics are quite coherent. Average of 0.85, 0.78, 0.92, 0.88.
Topic Coverage	100%	All 6 papers were assigned to a topic. Perfect coverage.

Metric	Value	Meaning
Gap Novelty	0.51 (51%)	Average gap score of $(0.54 + 0.58 + 0.42) / 3 = 0.51$. Moderate novelty.
Model Quality	65%	Composite: $0.86 \times 0.4 + 1.0 \times 0.3 + 0.51 \times 0.3 = 0.81$, but adjusted for dataset size. $\approx 65\%$.

Per-topic coherence:

- Medical Imaging: 85% (p1, p2 are similar)
- Federated Learning: 78% (p3, p5 overlap but p5 is multi-topic)
- Knowledge Graphs: 92% (p4 alone, very focused)
- Vision: 88% (p6 alone, classic paper)

Gap score distribution:

- 0.0–0.2: 0 gaps
- 0.2–0.4: 1 gap (g3)
- 0.4–0.6: 2 gaps (g1, g2)
- 0.6–0.8: 0 gaps
- 0.8–1.0: 0 gaps

What you see in the dashboard

Section 7 shows:

- The 4 summary cards
- 4 quality metric gauges
- A per-topic coherence chart
- A gap score histogram

Why this matters

This section is a diagnostic. It tells you:

- ✓ All papers were successfully grouped (100% coverage).
- ✓ Topics are cohesive (86% coherence).
- ✓ Gaps exist but are moderate in strength (51% novelty).
- ✓ Overall model quality is good (65%).

If any of these scores were low, you would know the analysis needs improvement.

Complete Flow Summary

Here is what happened to your 6 papers:

INPUT: 6 PDFs

↓

[Section 1] Parse: Extract all 6 papers successfully ✓


```
↓
[Section 2] Topic: Cluster into 4 topics
- Medical Imaging (p1, p2)
- Federated Learning (p3, p5)
- Knowledge Graphs (p4)
- Vision (p6)
↓
[Section 3] Gaps: Identify 3 gap pairs
- g2: Medical + Knowledge (0.58, strongest)
- g1: Medical + Federated (0.54)
- g3: Federated + Knowledge (0.42)
↓
[Section 4] Trends: Detect growth
- Rising: Medical, Federated
- Stable: Knowledge, Vision
↓
[Section 5] Map: Visualize layout
- 6 colored dots clustered by topic
- 3 red gap lines between centroids
↓
[Section 6] Chat: Answer questions
- Grounded in actual paper content
- Citations from top-ranked chunks
↓
[Section 7] Evaluation: Score the whole pipeline
- Coherence: 86%
- Coverage: 100%
- Novelty: 51%
- Quality: 65%
↓
OUTPUT: Actionable research insights
```

Key Takeaways

What the pipeline told us

1. Your dataset covers 4 distinct research areas.
2. Medical imaging and federated learning are emerging (rising trends).
3. The strongest research gap is between medical imaging and knowledge graphs (g2, 0.58 score).
4. All papers were successfully analyzed (100% coverage).
5. Topic coherence is good (86%), meaning topics are well-separated.

What you should do next

Based on these results, you might:

- **Investigate the gap:** Why aren't medical imaging and knowledge graphs combined? This could be a high-impact research direction.
- **Look at rising topics:** Medical imaging and federated learning are trending—both are active areas worth following.

- **Review topic assignments:** If any topic seems off (low coherence), check the papers manually.
 - **Ask the chatbot:** Use Section 6 to dive deeper into specific questions about the detected gaps and trends.
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Conclusion

ResearchLens turned your 6 papers into a structured analysis pipeline:

- Papers → Topics → Gaps → Trends → Map → Chat → Evaluation.

Each section builds on the previous one. The final evaluation tells you whether to trust the analysis. In this example, the results are solid (good coherence, full coverage, moderate novelty). The pipeline successfully identified actionable research gaps and emerging trends.

This is the power of automated research intelligence: see patterns in your corpus that would take days to find manually.